THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 1: **Exploration: Entering the World of Chemistry** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A cube of side 2.0 cm (+/- 0.1 cm). The percentage error in the volume is about: 1

(a) 15 per cent **(b)** 5 per cent **(c)** 10 per cent **(d)** 1.5 per cent

2. Which of the following is a good reason for conducting a pilot experiment? 1

(a) to produce final results
(b) to test the procedure and find problems
(c) to hide errors
(d) to save time by skipping the main experiment

3. Assertion (A): The mean of repeated readings is more reliable than a single reading. Reason (R): Random errors tend to cancel out on averaging. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

4. Which of the following represents the correct way to write 0.00045 in scientific notation? 1

(a) 4.5 x 10^4 **(b)** 0.45 x 10^-3 **(c)** 45 x 10^-5 **(d)** 4.5 x 10^-4

5. An instrument gives readings 10.1, 10.2, 10.1 when the true value is 12.0. The instrument is: 1

(a) accurate but not precise **(b)** precise but not accurate
(c) both accurate and precise **(d)** neither

6. Assertion (A): All scientific laws are guesses that have not been tested. Reason (R): Laws describe patterns observed repeatedly. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

7. If a model predicts 25 m but the experiment gives 24.8 m, the discrepancy is within the experimental error when the uncertainty is: 1

(a) 0.01 m **(b)** 0.001 m **(c)** 0.05 m **(d)** 0.5 m

8. An instrument gives readings 9.0, 11.0, 10.0, 12.0, 8.0 and the true value is 10.0. The instrument is: 1

(a) accurate on average but not precise **(b)** precise but not accurate
(c) both **(d)** neither

9. **Assertion (A):** A model can be useful in science even though it is simplified. 1
Reason (R): A scientific model is an exact copy of the real object or system.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** A scientific model, once accepted, can never be changed. 1
Reason (R): New evidence that disagrees with a model's predictions can lead scientists to modify or replace the model.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why do models of weather require supercomputers? 2

12. Why does the 'conservation of energy' act as a good check for the models and calculations? 2

13. Why is it unsafe to treat a hypothesis that fits many observations as 'proved'? 2

14. Why are 'zero errors' systematic errors while the reaction-time errors are random? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain why a hypothesis must be falsifiable and give an example of a hypothesis that is not. 3

16. Estimate how many drops of water make a litre if one drop is about 0.05 mL. 3

17. Explain why a result that cannot be repeated by other scientists is treated with caution. 3

OR

Find the percentage error: the density measured is 2.72 g/cm3 for aluminium whose accepted value is 2.70 g/cm3. Explain why the error is small. 3

18. Describe how scientists use dimensional analysis (unit check) to detect an error in a formula, using speed = distance x time as the wrong formula. 3

OR

A scientist notes that 'ice melts at 0 degrees Celsius' at normal pressure. Explain why the pressure is stated. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A bus from Jaipur to Agra covers 240 km in 4 hours. A second bus on the same route travels at a steady 25 m/s. Rohit wants to compare the two speeds, but they are given in different units. 5
(i) Find the average speed of the first bus in km/h.
(ii) Convert this speed to m/s.
(iii) Express 240 km in metres.
(iv) Convert the speed of the second bus to km/h and state which bus is faster and by how much in km/h.

20. Explain how scientists across the world cooperate using standard units, symbols and open data. Discuss the Mars Climate Orbiter failure (mix of pound-force and newton) as a case study. 5

OR

A group claims that a new substance is a better fuel because it heated 200 g of water by 40 degrees Celsius in 5 minutes. Describe the tests needed to compare it fairly with kerosene and the possible sources of error. 5

END OF QUESTION PAPER — Total: 40 Marks